Shree Rahul Education Society's (Regd.)

SHREE L. R. TIWARI COLLEGE OF ENGINEERING

An Autonomous Institute Affiliated to University of Mumbai,

Approved by AICTE & DTE, Maharashtra State | NAAC Accredited, NBA Accredited Program | ISO 9001:2015 Certified | DTE Code No: 3423 | Recognized under Section 2(f) of the UGC Act 1956 | Minority Status (Hindi Linguistic)

DEPARTMENT OF COMPUTER ENGINEERING

CML610 Cloud Computing Lab

Sixth Semester, 2025-2026 (Even Semester)

Name of Student : Ayan Parkar

Roll No. : 48

Division : A

Batch : A2

Day / Session :

Venue :

Experiment No. : 6

Title of Experiment : Study and Implement of Storage as a Service

Date of Conduction :

Date of Submission :

Particulars	Max. Marks	Marks Obtained
Preparedness and Efforts(PE)	**3**	
Knowledge of tools(KT)	**3**	
Debugging and results(DR)	**3**	
Documentation(DN)	**3**	
Punctuality & Lab Ethics(PL)	**3**	
Total	**15**	

Grades – Meet Expectations (3 Marks), Moderate Expectations (2 Marks), Below Expectations (1 Mark)

Checked and Verified by

Name of Faculty : Pravin Jangid

Signature :

Date :

Experiment No 6

Aim: Study and Implement of Storage as a Service

Theory:

1. Prepare a detailed study of Storage as a Service
Storage as a Service (STaaS) is a cloud computing model where a service provider rents scalable storage infrastructure to an organization on a subscription basis. This approach replaces traditional on-premises capital expenditures with an operational expense model, allowing businesses to dynamically scale their storage needs without managing physical hardware.

Cloud providers typically offer three primary storage architectures to accommodate different workload requirements, performance needs, and data structures. Block storage provides low-latency access for databases, file storage organizes data in hierarchical folders for shared access, and object storage handles massive unstructured datasets using metadata.

Adopting STaaS eliminates the upfront costs and maintenance burden of physical storage arrays, converting unpredictable hardware investments into predictable operational expenses. Furthermore, providers guarantee high availability and performance through Service Level Agreements (SLAs), while offering seamless scalability to accommodate sudden data growth or technological advancements.

Securing STaaS requires robust Identity and Access Management (IAM) systems incorporating multi-factor authentication and role-based access control to protect data in multi-tenant environments. Organizations must also navigate complex compliance landscapes, ensuring their chosen provider adheres to strict data residency and auditing standards mandated by frameworks like GDPR and HIPAA.

STaaS is foundational for advanced machine learning workflows, providing the immense capacity and high throughput required to process large datasets and train massive language models. In these pipelines, object and high-performance file storage are frequently used for preparing training data, loading model weights, and repeatedly saving inference checkpoints without bottlenecking the compute nodes.

The STaaS market is dominated by hyperscalers like Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP), which offer comprehensive ecosystems of block, file, and object services. Additionally, specialized enterprise providers like Pure Storage, IBM, and NetApp deliver high-performance, AI-optimized storage solutions for complex on-premises and hybrid cloud deployments.

2. Advantages and Limitations of StaaS for S3 & S3 Glacier Service

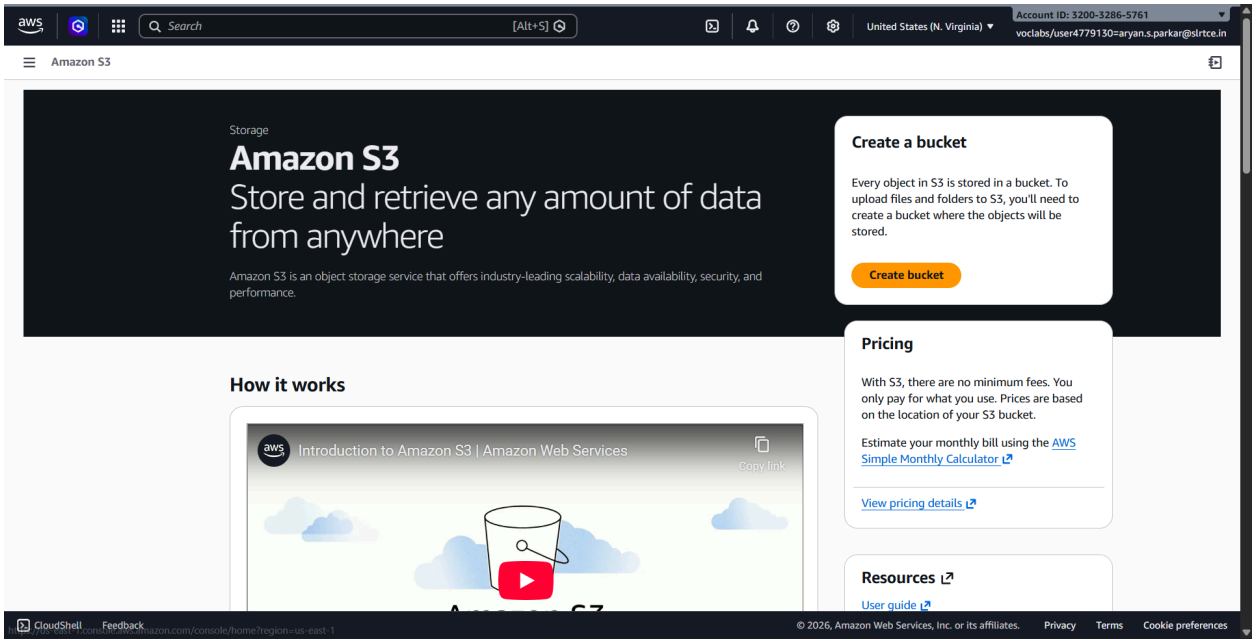
Aspect	Amazon S3	Amazon S3 Glacier
Scalability	Scales automatically to exabyte levels with no manual provisioning	Also scales without upfront provisioning; a single archive can hold up to 40TB
Durability	Designed for 99.999999999% data durability	Same 99.999999999% durability; built for long-term archival reliability
Cost Efficiency	Pay-as-you-go; eliminates need for physical infrastructure investment	Significantly lower per-GB cost vs S3 Standard
Data Access Speed	Millisecond latency; supports real-time high-throughput workloads	Retrieval times range from minutes to hours depending on tier (Instant, Flexible, Deep Archive)
Security	Millisecond latency; supports real-time, high-throughput workloads	Supports SSL for data in transit and server-side encryption at rest
Use Case	Web hosting, analytics, active datasets, user-generated content	Backup, compliance archiving, disaster recovery for rarely accessed data
Retrieval Cost	No retrieval fees on Standard; request-based pricing	Retrieval fees apply (except bulk); cost increases with faster retrieval tiers
Minimum Storage Duration	No minimum; billed per actual usage period	Minimum storage charge applies (40 KB per object minimum)
Infrastructure Control	Limited control over underlying hardware and infrastructure	Same; users cannot manage physical storage nodes or retrieval pipelines
Long-Term Cost	Subscription costs may exceed on-premises investment over very long periods	Lower than S3 Standard long-term, but retrieval + management fees can accumulate

Activity:

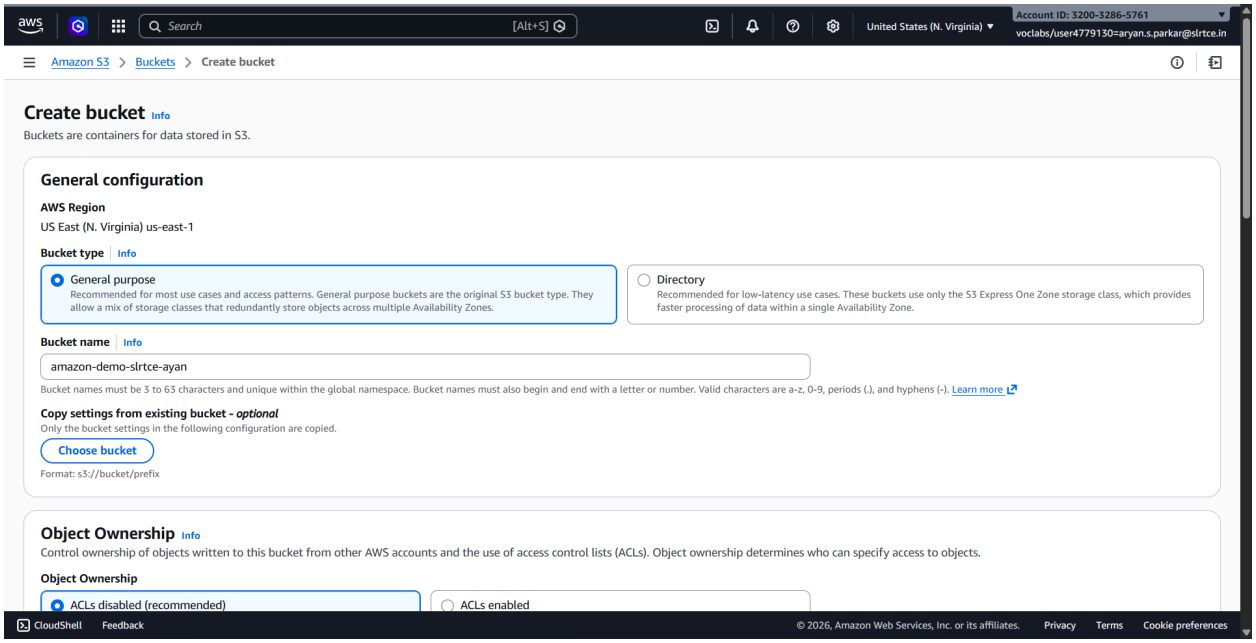
Step 1: Log in to AWS Console

Step 2: Search S3 or click on the following link: console.aws.amazon.com/s3/get-started

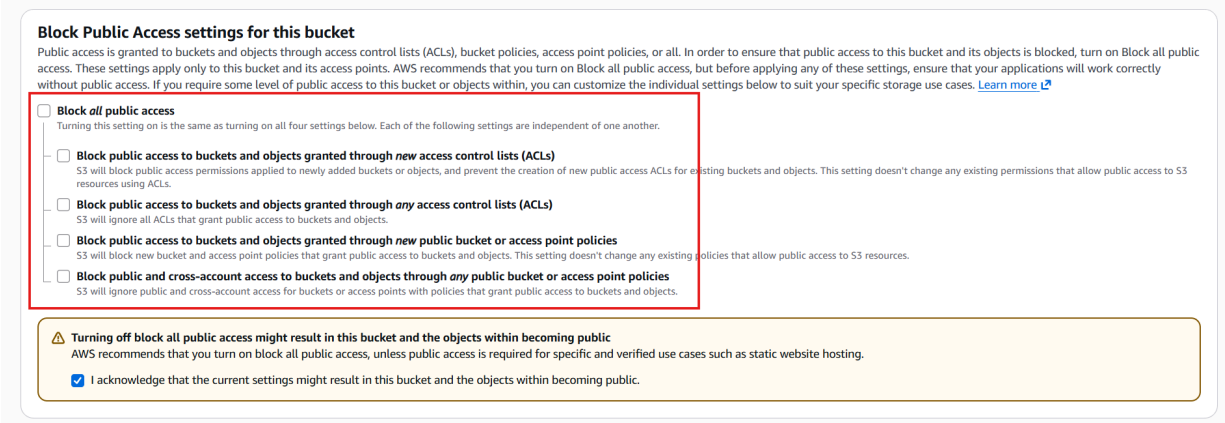
Step 3: Click on Create bucket



Step 4: Configure the bucket

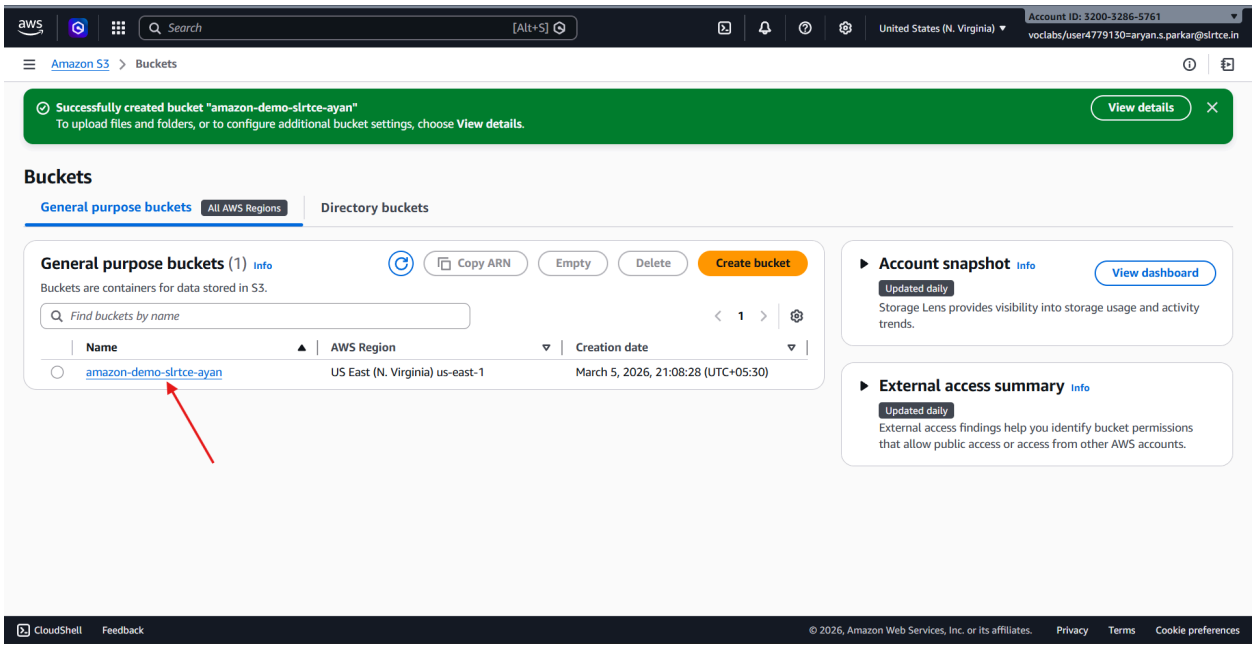


Step 5: Make sure that these options are unchecked

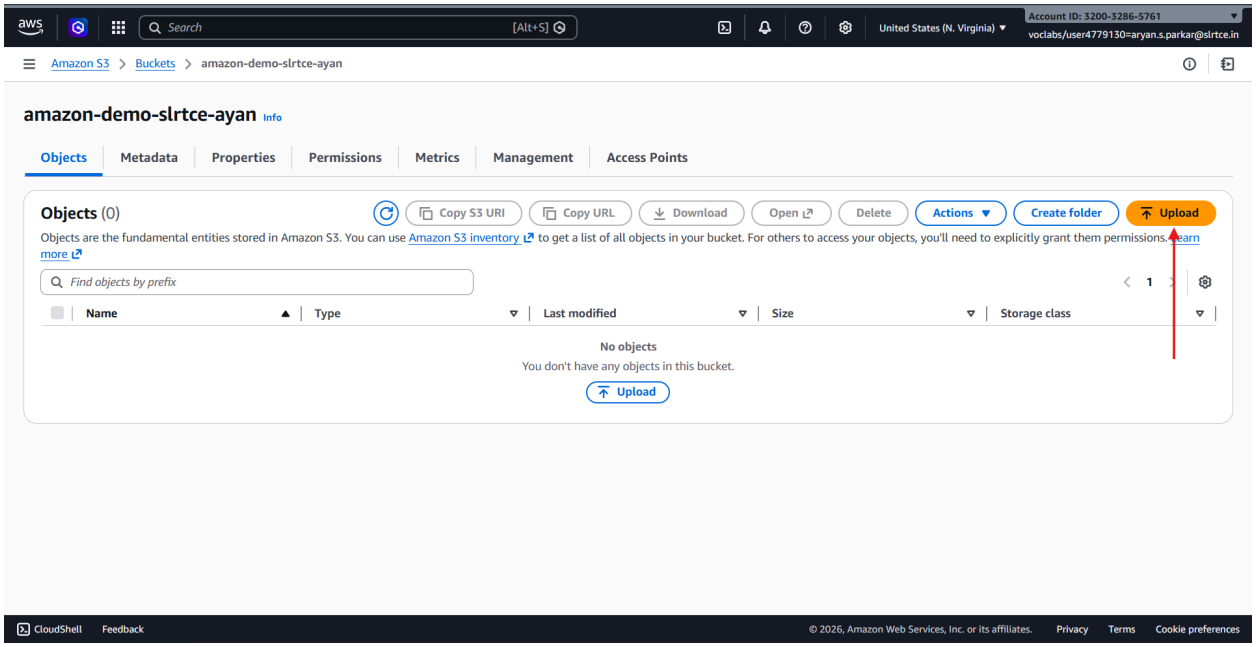


Step 6: Create Bucket

Step 7: Upon creation of the bucket, click on the newly created bucket

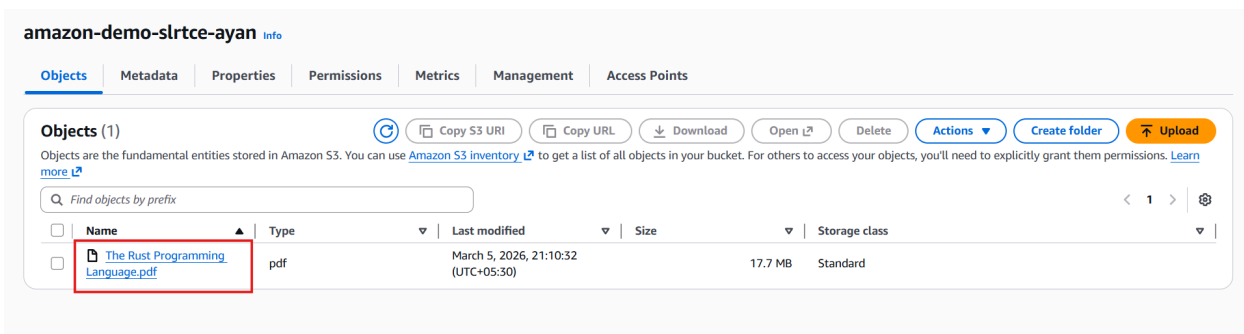


Step 8: Click on the Upload button to upload any file on the bucket

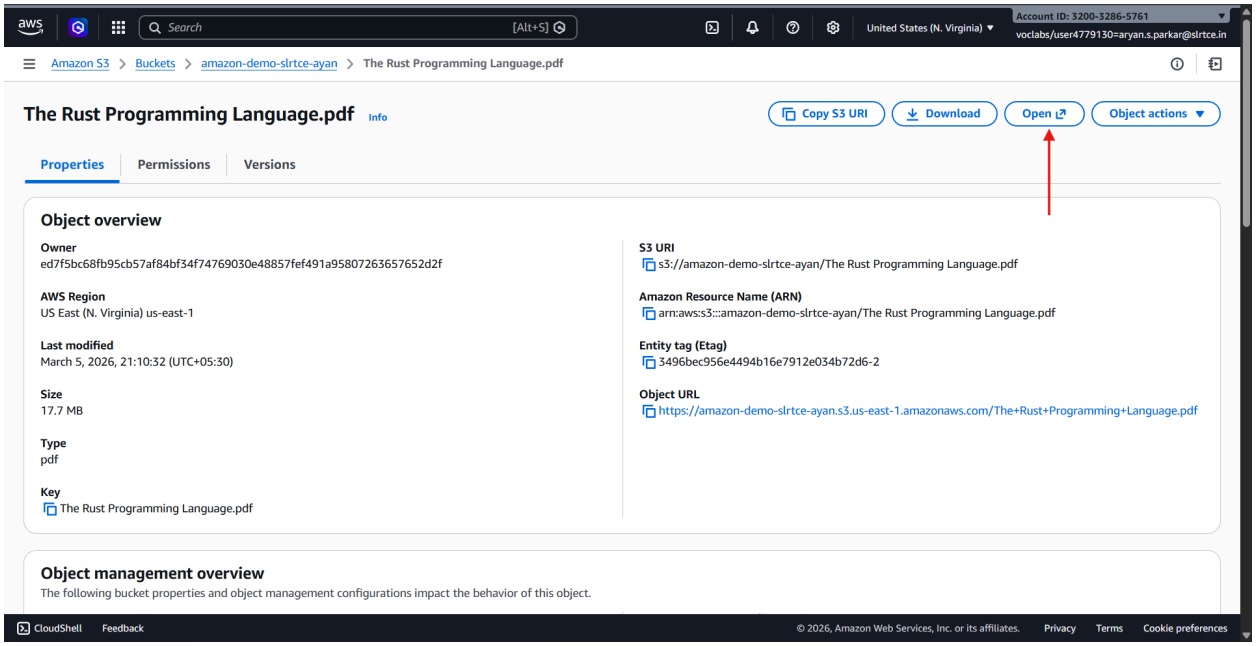


Step 9: Add the file(s) and click on Upload

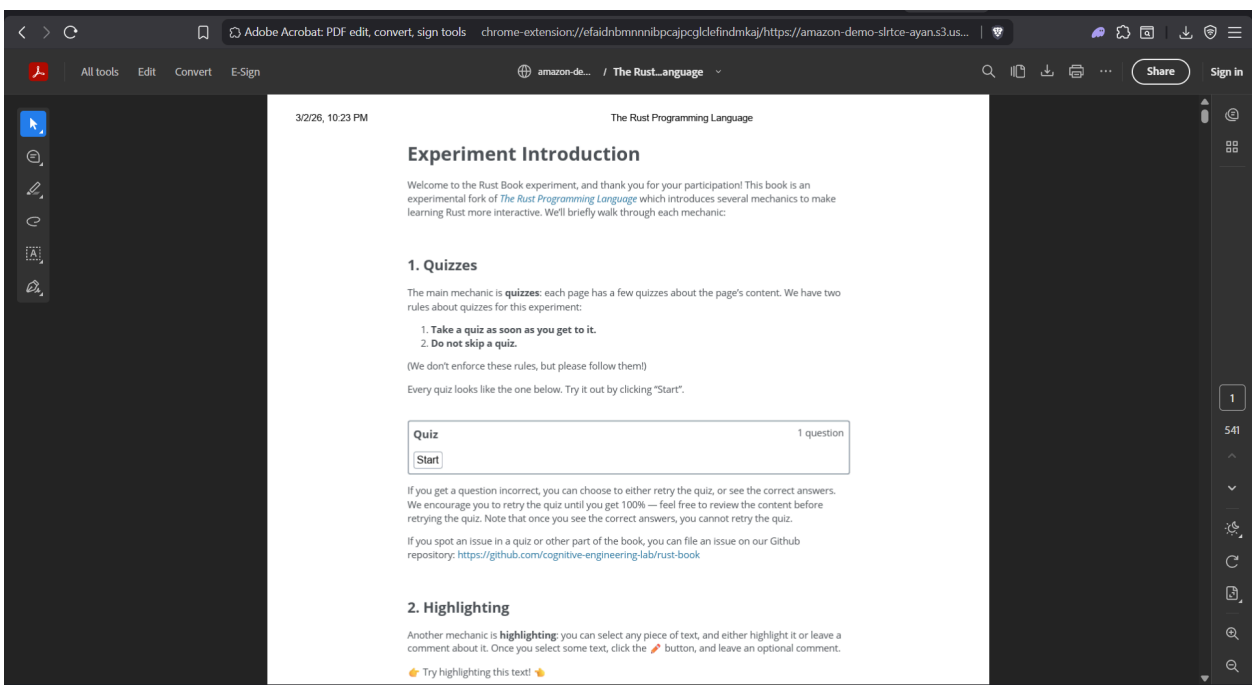
Step 10: The uploaded file will show up like this



Step 11: Click on the file -> Click on the Open button to then open your file that has been uploaded to a cloud storage



The file will then open up in your browser



Conclusion:

Hence, we successfully studied the Storage as a Service (STaaS) and deployed a S3 bucket and hosted our files online.